

PROJECT DEVELOPMENT

POCKETSMART AI — Smart Budget & Recommendation Assistant

Expanded phase evidence with detailed wording, diagrams, workflow, and project-focused explanation.

Implementation overview

The development phase turns the design into application code. The documented project uses Python and FastAPI for the backend, Jinja2 for server-rendered templates, HTML/CSS for the interface, Pydantic for data models, and a Gemini API utility for AI-generated recommendations. The implementation is organized so each major responsibility can be traced to a file, route, model, template, or utility.

Backend development

app.py coordinates routes, templates, planner requests, and recommendation flow. auth.py handles password hashing, authentication, access tokens, and protected-user checks. models.py defines structured inputs for registration and the Home, Party, and Jewelry planners. gemini_utils.py connects planner context to Gemini and formats structured recommendation data. Together these pieces form the core backend workflow.

Frontend and recommendation experience

The frontend contains dedicated templates for registration, login, dashboard, planners, recommendation pages, and History. The user moves from structured input to a generated result instead of manually interpreting raw AI text. Recommendation pages can show budget summaries, category-level suggestions, descriptions, estimated prices, quantities, shopping references, and additional guidance according to the planner.

Visual Model

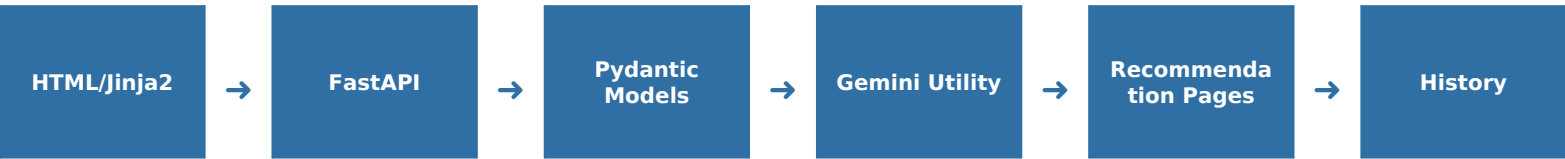


Figure — Phase visual model

Process Flow



Figure — Detailed workflow / roadmap

PHASE 5 — DETAILED CONTENT

PocketSmart AI connects practical planning with a structured recommendation workflow. The following section expands the phase in clear project language so the document has sufficient explanatory depth for academic/project review.

Project Explanation

A strong budget-planning assistant should reduce uncertainty without hiding the reasoning context behind the result. In PocketSmart AI, the user supplies the information that matters for the selected category, including budget and relevant preferences. The application then processes that context and presents a structured recommendation. This structure makes the output easier to read, compare, and explain during a project demonstration. The project also demonstrates how a common architecture can support different use cases. Home Interior planning focuses on rooms, furniture, lighting, fans, and dining requirements. Party planning focuses on guests, event type, venue type, catering, decoration, and entertainment. Jewelry planning focuses on occasion, style preferences, budget, and optional outfit-image context. Although the inputs differ, the overall application pattern remains consistent.

Key Concepts

Component	Development role
app.py	Main FastAPI application and route orchestration.
auth.py	Authentication and token-related operations.
models.py	Pydantic/data model definitions.
gemini_utils.py	Gemini API interaction and structured recommendation handling.
Templates	User-facing forms, results, dashboard, and History.

Roadmap / Mind-Map View



Figure — Additional project roadmap visual

Project Evidence

The supplied project documentation includes interface and workflow evidence. A representative screenshot is included here to connect the written explanation with the implemented application.

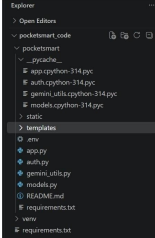
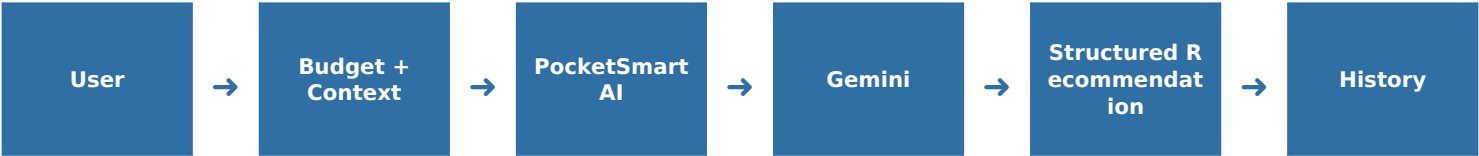


Figure — Representative PocketSmart AI project evidence

PHASE SUMMARY

Phase 5 contributes to the overall PocketSmart AI lifecycle by making the project easier to understand, build, validate, or present. The emphasis in this phase is on clear communication: the reader should be able to see what the system is doing, why the step matters, and how it connects to the final budget-aware recommendation workflow.

End-to-End Connection



Project Reflection

PocketSmart AI is designed around a practical user problem: making planning decisions while respecting a defined budget. The project brings together web development, authentication, structured data, Generative AI, visual presentation, and History. This combination makes the application a useful example of how a software project can move from a simple idea to a demonstrable end-to-end system. The documentation should therefore communicate not only the features, but also the reasoning, workflow, and evidence behind them.

Closing Note

The phase is complete when its purpose, workflow, supporting evidence, and connection to PocketSmart AI are understandable to a reviewer without requiring them to infer missing steps. The expanded paragraphs and diagrams in this version are intended to make that explanation clearer and more professional.